DT.

selection Sort

selection short is a simple algorithm that repeatedly selects the smallest element from the unsorted part of the array and places it at its correct position.

After every pass, one element is placed in its correct sorted position.

Working Principle

1. Assume the first element in the smallest
2. compare it with all remaining elements
3. Find the actual smallest element
4. Swap it with the first element
5. Move to the next position and repeat the same process
6. continue untill the entire array is sorted.

Algorithm

Step 1 Start from the index

i=0 ;

Step-2 Assume the current index contains the minimum element

min Index = i;

DT.

step-3 Traverse the remaining array

for (J = i+1 , J<n ; J++)

step-4 If a smaller element is found

arr[J] < arr [minIndex]

update

minIndex = J;

step-5 After finding minimum element, swap it with the current element

swap (arr [minIndex], arr [i]);

Program :-

```
void Selection Sort (vector<int> & arr, int n)
{
   for (int i=0 ; i<n-1 ; i++ ){
      int minIndex =i;
      For (int j = i+1 , J<n; J++)
      {
          if (arr [j] < arr ([minIndex]))
              minindex = J;
      }
      swap (arr [minIndex], arr [i]);
   }
}
```

Spiral

DT.

Dry run

Array :- 13 46 24 52 20 9

Pass 1
minimum 9

13 46 24 52 20 9

swap with first element

9 46 24 52 20 13

Pass 2
unsorted 46 24 52 20 13
minimum = 13
swap

9 13 24 52 20 46

Pass -3
minimum = 20

9 13 20 52 24 46

Pass -4.
minimum = 24

9 13 20 24 52 46

Pass - 5 minimum → 56

9 13 20 24 46 52.

Spiral